

Qatar-9b

R-2321

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-9_250519 · created 2026-08-18T00:57:35+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460814.7182 +/- 0.0093 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.199 +/- 0.057
Transit depth (Rp/Rs)^2	3.95 +/- 2.25 [%]
Orbital Inclination (inc)	84.28 +/- 1.66
Airmass coefficient 1 (a1)	0.839 +/- 0.022
Airmass coefficient 2 (a2)	0.15 +/- 0.13
Scatter in the residuals of the lightcurve fit is	2.56 %
Best Comparison Star	#3 - [340, 347]
Optimal Method	PSF photometry
Transit Duration (day)	0.043 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.199 $\pm$ 0.057 vs 0.1489 (deviation 0.8 σ)
Duration fit vs published	0.043 d vs 0.0779 d (deviation 2.9 σ)
Mid-time O-C	-0.4 min
Depth SNR	3.2
Residual scatter	2.56 %

Light curve

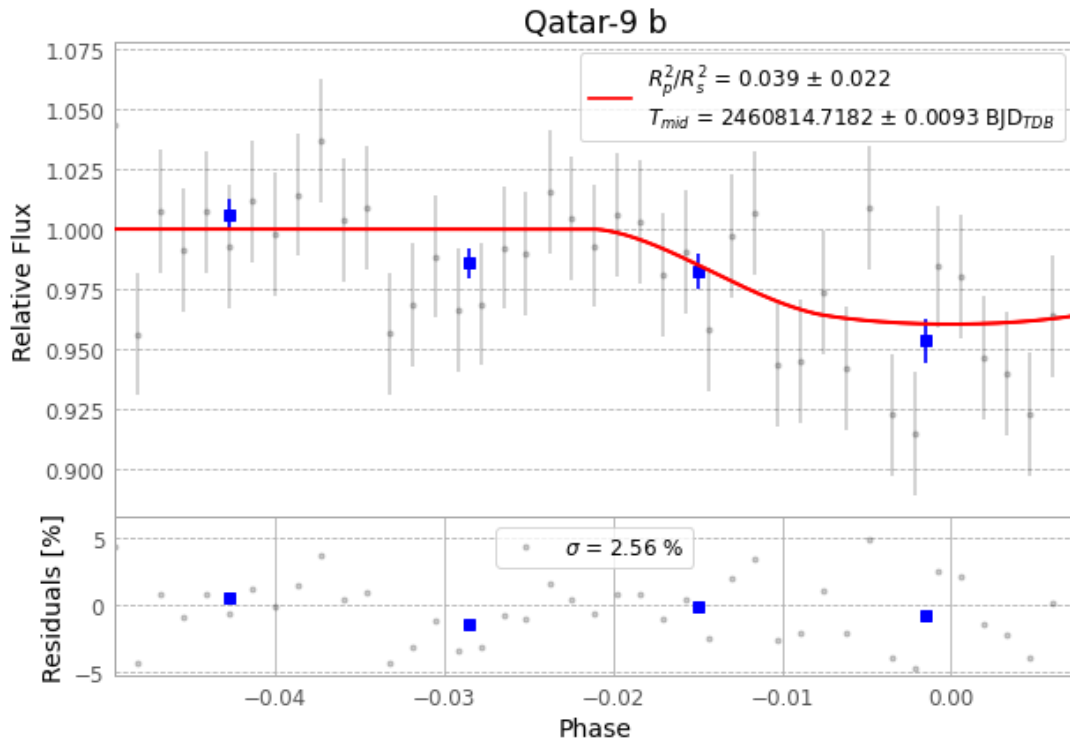


Figure 1 — FinalLightCurve_Qatar-9 b_2025-05-18.png (SHA-256 536fe1756901afb0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [364, 280], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 af03d436aa537e7a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

43 FITS frames (28 MB), Qatar-9250519032416.FITS → Qatar-9250519053015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-9-250519.log (c88ee59df4dd...), FinalLightCurve_Qatar-9 b_2025-05-18.png (536fe1756901...), FinalLightCurve_Qatar-9 b_2025-05-18.csv (069019b8ae1b...)

Compute resources

Wall time 82.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 00:57Z.